

**SIES COLLEGE OF ARTS, SCIENCE & COMMERCE
(AUTONOMOUS)
SION(W), MUMBAI-22**

DEPARTMENT OF INFORMATION TECHNOLOGY

MSc(IT), SEMESTER IV

Practical Journal

For the Subject

Natural Language Processing

Submitted by

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Department of Information Technology

CERTIFICATE

This is to certify that **Neerajkumar Satyaprakash Gupta**, Seat No. **SMSC2425224** has successfully completed the necessary course of experiments in the subject of **Natural Language Processing** with the subject code **SIPITCCP622** during the academic year 2024 – 2025 complying with the requirements of University of Mumbai, for the course of **MSc.in Information Technology Part 2 [Semester-4]**

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Date:

College Seal

INDEX

SR NO.	PRACTICAL NAME	SIGNATURE
1	Write a Program to Perform Tokenization over word and sentence on English and Hindi Text.	
2	Write a Program to identify Stopwords in a given sentence in English.	
3	Write a program to perform Stemming and Lemmatization for English Text.	
4	Write a program to segregate Part of Speech (POS Tagging) for English Text.	
5	Write a program to perform Name Entity Recognition (NER) & Chunking on English Text.	
6	Write a program to perform WordNet & also check Word Similarity on English Text.	
7	Write a program to implement word cloud of English Text.	
8	Write a program to process Text Summarization.	
9	Write a program to implement Word2Vec on Wikipedia Articles and finding the similarity between the words.	
10	Write a program to Train a model for Movie Review Classification using NLP Techniques.	

✓ Practical No 1

Write a program to perform tokenization over word and sentence on English and Hindi Text

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
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```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
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[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
# Tokenization
# import the Library
from nltk.tokenize import word_tokenize
from nltk.tokenize import sent_tokenize
```

```
# Importing data
dataset = """ Hello Mr. Watson, how are you doing today?
The weather is awesome.
The garden is green.
We should go out for a walk
The sky is pinkish-blue.
You shouldn't eat cardboard. """
```

```
# Tokenization the sentences
print(sent_tokenize(dataset))
for i in sent_tokenize(dataset):
    print(i)
```

```
[' Hello Mr. Watson, how are you doing today?', 'The weather is awesome.', 'The garden is green.', 'We should go out for a walk\nThe  
Hello Mr. Watson, how are you doing today?  
The weather is awesome.  
The garden is green.  
We should go out for a walk  
The sky is pinkish-blue.  
You shouldn't eat cardboard.'
```

```
from nltk.tokenize import word_tokenize
print("word_tokenize", word_tokenize(dataset))
```

```
word_tokenize ['Hello', 'Mr.', 'Watson', ',', 'how', 'are', 'you', 'doing', 'today', '?', 'The', 'weather', 'is', 'awesome', '.', '1
```

```
from nltk.tokenize import TreebankWordTokenizer
# tokenizers work by separting the words using punctuation and spaces
tokenizer = TreebankWordTokenizer()
print("TreebankWordtokenizer", tokenizer.tokenize(dataset))
```

```
TreebankWordtokenizer ['Hello', 'Mr.', 'Watson', ',', 'how', 'are', 'you', 'doing', 'today', '?', 'The', 'weather', 'is', 'awesome.'
```

```
from nltk.tokenize import word_tokenize
text = """ नमस्ते श्रीमान वाटसन,  
आप आज कैसे हैं?  
मौसम शानदार है।  
बगीचा हरा है।  
हमें टहलने के लिए बाहर जाना चाहिए।  
आसमान गुलाबी-नीला है।  
आपको कार्डबोर्ड नहीं खाना चाहिए। """
print("Word_tokenize", word_tokenize(text))
```

```
Word_tokenize ['नमस्ते', 'श्रीमान', 'वाटसन', ',', 'आप', 'आज', 'कैसे', 'हैं', '?', 'मौसम', 'शानदार', 'है।', 'बगीचा', 'हरा', 'है।', 'हमें', 'टह
```

```
from nltk.tokenize import sent_tokenize
print("Sentence_tokenize", sent_tokenize(text))
```

```
Sentence_tokenize [' नमस्ते श्रीमान वाटसन,\nआप आज कैसे हैं?', 'मौसम शानदार है।\nबगीचा हरा है।\nहमें टहलने के लिए बाहर जाना चाहिए।\nआसमान गुलाबी-न
```

```
from nltk.tokenize import WordPunctTokenizer
# separates the punctuation from the word.
tokenizer = WordPunctTokenizer()
print("WordPunctTokenizer", tokenizer.tokenize(text))
```

```
WordPunctTokenizer ['नमस', '्', 'त', 'े', 'श', 'र्', 'र', 'ी', 'म', 'ा', 'न', 'व', 'ा', 'टसन', ',', 'आप', 'आज', 'क', 'ै', 'स', 'े'
```

```
from nltk.tokenize import TreebankWordTokenizer
# tokenizers work by seperating the words using punctuation and spaces
tokenizer = TreebankWordTokenizer()
print("TreebankWordTokenizer", tokenizer.tokenize(text))
```

```
TreebankWordTokenizer ['नमस्ते', 'श्रीमान', 'वाटसन', ',', 'आप', 'आज', 'कैसे', 'हैं', '?', 'मौसम', 'शानदार', 'है।', 'बगीचा', 'हरा', 'है।', 'ह
```

Start coding or [generate](#) with AI.

✓ Practical No 2

Write a Program to identify Stopwords in a given sentence in English.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
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Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
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[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
dataset = """ English is a very important asset when seeking employment in Guatemala. Help end the language barrier by being proficient
```

```
stop_words = set(stopwords.words('english'))
print(stop_words)
```

```
{'being', 'we', 'his', 'itself', 't', "weren't", 'when', 'hers', 'off', 'aren', 'until', 'an', 'they', 'but', "they'll", "i've", 'st
```

```
print("Total count of stopwords:", len(dataset))
```

```
Total count of stopwords: 293
```

```
words = word_tokenize(dataset)
print(words)
```

```
['English', 'is', 'a', 'very', 'important', 'asset', 'when', 'seeking', 'employment', 'in', 'Guatemala', '.', 'Help', 'end', 'the',
```

```
print("total words: ", len(words))
```

```
total words: 49
```

```
filtered_sentence=[]
for w in words:
    if w not in stop_words:
        filtered_sentence.append(w)
print(filtered_sentence)
print()
print("After removing stopwords", len(filtered_sentence))
```

```
['English', 'important', 'asset', 'seeking', 'employment', 'Guatemala', '.', 'Help', 'end', 'language', 'barrier', 'proficient', 'Er
```

After removing stopwords 31

```
filtered_sentence1=[]
for w in words:
    if w in stop_words:
        filtered_sentence1.append(w)
print(filtered_sentence1)
print()
print("Exsiting stopwords", len(filtered_sentence1))
```

```
['is', 'a', 'very', 'when', 'in', 'the', 'by', 'being', 'in', 'which', 'can', 'to', 'in', 'such', 'as', 'and', 'more', 'your']
```

Exsiting stopwords 18

✓ Practical No 3

Write a program to perform Stemming and Lemmatization for English Text

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
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Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
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[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
# import these modules
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize

ps = PorterStemmer()
```



```
# choose some words to be stemmed
words = ["program", "programs", "programer", "programming", "programmers"]

for w in words:
    print(w, " : ", ps.stem(w))
```

```
↵ program : program
   programs : program
   programer : program
   programming : program
   programmers : programm
```

```
# choose some words to be stemmed
sentence = "Programmers program with programming languages"
words = word_tokenize(sentence)
for w in words:
    print(w, " : ", ps.stem(w))
```

```
↵ Programmers : programm
   program : program
   with : with
   programming : program
   languages : languag
```

```
from nltk.stem import PorterStemmer
e_words = ["wait", "waiting", "waited", "waits"]
ps = PorterStemmer()
for w in e_words:
    rootWord = ps.stem(w)
    print(rootWord)
```

```
↵ wait
   wait
   wait
   wait
```

```
from nltk.stem import PorterStemmer
from nltk.tokenize import sent_tokenize, word_tokenize
```

```
sentence = "He was waiting for the boat at the dock, watching as the waves gently lapped against the pier. The sun was beginning to set,
words = word_tokenize(sentence)
ps = PorterStemmer()
for w in words:
    rootWord = ps.stem(w)
    print(w,rootWord)
```

```
↵ He he
   was wa
   waiting wait
   for for
   the the
   boat boat
   at at
   the the
   dock dock
   , ,
   watching watch
   as as
   the the
   waves wave
   gently gentli
   lapped lap
   against against
   the the
   pier pier
   . .
   The the
   sun sun
   was wa
   beginning begin
   to to
   set set
   , ,
   casting cast
   a a
   warm warm
   golden golden
   glow glow
   over over
   the the
   water water
   , ,
   making make
   the the
   moment moment
```

```

even even
more more
picturesque picturesque
and and
serene seren
. .

```

```

# create an object of class PorterStemmer
from nltk.stem import PorterStemmer
from nltk.stem import LancasterStemmer
porter = PorterStemmer()
lancaster=LancasterStemmer()

# provide a word to be stemmed
print("Porter Stemmer: - ")
print(porter.stem("cats"))
print(porter.stem("trouble"))
print(porter.stem("troubling"))
print(porter.stem("troubled"))
print()

print("Lancaster Stemmer: -")
print(lancaster.stem("cats"))
print(lancaster.stem("trouble"))
print(lancaster.stem("troubling"))
print(lancaster.stem("troubled"))

```

```

↔ Porter Stemmer: -
cat
troubl
troubl
troubl

Lancaster Stemmer: -
cat
troubl
troubl
troubl

```

```

# A list of words to be stemmed

```

```

word_list = ["friend", "friendship", "friends", "friendships", "companionship", "ally", "allies", "companions", "acquaintances", "bond",
print("{0:20}{1:20}{2:20}".format("Word", "Porter Stemmer", "Lancaster Stemmer"))

```

```

for word in word_list:
    print("{0:20}{1:20}{2:20}".format(word, porter.stem(word), lancaster.stem(word)))

```

```

↔ Word          Porter Stemmer  Lancaster Stemmer
friend          friend         friend
friendship     friendship    friend
friends        friend        friend
friendships    friendship    friend
companionship  companionship comp
ally           alli          al
allies         alli          al
companions     companion    comp
acquaintances  acquaint      acquaint
bond           bond          bond
relationship    relationship  rel
connections    connect      connect
affinity       affin        affin
support        support      support
trust          trust        trust
affection      affect       affect
loyalty        loyalty      loyal
comfort        comfort      comfort
companionship  companionship comp
kinship        kinship      kin
partnership    partnership  partn
closeness      close        clos
rapport        rapport      rapport
understanding  understand   understand
solidarity     solidar      solid
association    associ      assoc
fellowship     fellowship  fellow
comradeship    comradeship comrad
mutuality      mutual      mut
supportiveness support      support
amity          amiti        am
camaraderie    camaraderi   camaradery
intimacy       intimaci     intim
sympathy       sympathi    sympathy
empathy        empathi     empathy
trustworthiness trustworthi  trustworthy
faithfulness   faith        faith
kindness       kind         kind

```

generosity
togetherness

generos
togeth

generos
togeth

```
sentence = "Pythoners are very intelligent and work very pythonly and now they are pythoning their way to success."  
porter.stem(sentence)
```

→ 'pythoners are very intelligent and work very pythonly and now they are pythoning their way to success.'

```
# stemming  
import nltk  
from nltk.stem.porter import PorterStemmer  
porter_stemmer = PorterStemmer()  
text = "studies studying cries cry"  
tokenization = nltk.word_tokenize(text)  
for w in tokenization:  
    print("Stemming for {} is {}".format(w, porter_stemmer.stem(w)))
```

→ Stemming for studies is studi
Stemming for studying is studi
Stemming for cries is cri
Stemming for cry is cri

```
# Lemmatization  
import nltk  
from nltk.stem import WordNetLemmatizer  
wordnet_lemmatizer = WordNetLemmatizer()  
text = "studies studying cries cry"  
tokenization = nltk.word_tokenize(text)  
for w in tokenization:  
    print("Lemma for {} is {}".format(w, wordnet_lemmatizer.lemmatize(w)))
```

→ Lemma for studies is study
Lemma for studying is studying
Lemma for cries is cry
Lemma for cry is cry

✓ Practical No 4

Write a program to segregate Part Of Speech (POS Tagging) for english Text.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
[nltk_data] | Downloading package tagsets_json to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package tagsets_json is already up-to-date!
[nltk_data] | Downloading package timit to /root/nltk_data...
[nltk_data] | Package timit is already up-to-date!
[nltk_data] | Downloading package toolbox to /root/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!
[nltk_data] | Downloading package treebank to /root/nltk_data...
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_tagset is already up-to-date!
[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
import nltk
from nltk.tokenize import wordpunct_tokenize
from nltk.tag import pos_tag
```

```
dataset = """The Taj Mahal, located in Agra, India, is one of the most iconic and beautiful monuments in the world; it was commissioned
```

```
# Tokenize the data
new_data = wordpunct_tokenize(dataset)
print(new_data)
```

```
['The', 'Taj', 'Mahal', ',', 'located', 'in', 'Agra', ',', 'India', ',', 'is', 'one', 'of', 'the', 'most', 'iconic', 'and', 'beautif
```

```
# Apply the POS Tagging
pos_tag(new_data)
```

```
[('The', 'DT'),
 ('Taj', 'NNP'),
 ('Mahal', 'NNP'),
 (',', ','),
 ('located', 'VBN'),
 ('in', 'IN'),
 ('Agra', 'NNP'),
 (',', ','),
 ('India', 'NNP'),
 (',', ','),
 ('is', 'VBZ'),
 ('one', 'CD'),
 ('of', 'IN'),
 ('the', 'DT'),
 ('most', 'RBS'),
 ('iconic', 'JJ'),
 ('and', 'CC'),
 ('beautiful', 'JJ'),
 ('monuments', 'NNS'),
 ('in', 'IN'),
 ('the', 'DT'),
 ('world', 'NN'),
 (';', ';'),
 ('it', 'PRP'),
 ('was', 'VBD'),
 ('commissioned', 'VBN'),
 ('by', 'IN'),
 ('Mughal', 'NNP'),
 ('Emperor', 'NNP'),
 ('Shah', 'NNP'),
 ('Jahan', 'NNP'),
 ('in', 'IN'),
 ('memory', 'NN'),
 ('of', 'IN'),
 ('his', 'PRP$'),
 ('beloved', 'JJ'),
 ('wife', 'NN'),
 (',', ','),
 ('Mumtaz', 'NNP'),
 ('Mahal', 'NNP'),
 (',', ','),
 ('and', 'CC'),
 ('is', 'VBZ'),
 ('a', 'DT'),
 ('symbol', 'NN'),
 ('of', 'IN'),
 ('eternal', 'JJ'),
 ('love', 'NN'),
 ('.', '.'),
 ('The', 'DT'),
 ('construction', 'NN'),
 ('of', 'IN'),
 ('the', 'DT'),
 ('Taj', 'NNP'),
 ('Mahal', 'NNP'),
 ('began', 'VBD'),
 ('in', 'IN'),
 ('1632', 'CD'),
```

```
# Tag Set
nltk.help.upenn_tagset()
```





✓ Practical No 5

Write a program to perform Name Entity Recognition(NER) & Chunking on English Text.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
[nltk_data] | Downloading package tagsets_json to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package tagsets_json is already up-to-date!
[nltk_data] | Downloading package timit to /root/nltk_data...
[nltk_data] | Package timit is already up-to-date!
[nltk_data] | Downloading package toolbox to /root/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!
[nltk_data] | Downloading package treebank to /root/nltk_data...
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_tagset is already up-to-date!
[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
from nltk.tokenize import word_tokenize
from nltk.tag import pos_tag
from nltk.chunk import ne_chunk
```

```
dataset="""Abraham Lincoln was an American statesman, lawyer, and the 16th president of the United States, serving from 1861 until his i
```

↳ [('Abraham', 'NNP'), ('Lincoln', 'NNP'), ('was', 'VBD'), ('an', 'DT'), ('American', 'JJ'), ('statesman', 'NN'), ('.', ',', ','), ('lawyer', 'NN')]

→ (S
(PERSON Abraham/NNP)
(PERSON Lincoln/NNP)
was/VBD
an/DT
(GPE American/JJ)
statesman/NN
,/
lawyer/NN
,/
and/CC
the/DT
16th/CD
president/NN
of/IN
the/DT
(GPE United/NNP States/NNPS)
,/
serving/VBG
from/IN
1861/CD
until/IN
his/PRP\$
assassination/NN
in/IN
1865/CD
./.
He/PRP
led/VBD
the/DT
nation/NN
through/IN
the/DT
Civil/NNP
War/NNP
,/
preserved/VBD
the/DT
(ORGANIZATION Union/NNP)
,/
abolished/VBD
slavery/NN
,/
strengthened/VBD
the/DT
federal/JJ
government/NN
,/
and/CC
modernized/VBD
the/DT
(GPE U.S./NNP)
economy/NN
./.)

→ ['Taj', 'Mahal', 'is', 'one', 'of', 'the', 'world', "'s", 'most', 'celebrated', 'structures', 'in', 'the', 'world', '.', 'It', 'is',

```
↳ [ ('Taj', 'NNP'), ('Mahal', 'NNP'), ('is', 'VBZ'), ('one', 'CD'), ('of', 'IN'), ('the', 'DT'), ('world', 'NN'), ('s', 'POS'), ('most
```



```
# defining sequence of chunk
sequence_chunk = """
chunk:
{<NNPS>+}
{<NNP>+}
{<NN>+}
"""

# creating object with regular expression
chunk=RegexpParser(sequence_chunk)

chunk_result=chunk.parse(postagging)
print(chunk_result)
```

```
most/RBS
celebrated/JJ
structures/NNS
in/IN
the/DT
(chunk world/NN)
./
It/PRP
is/VBZ
a/DT
stunning/JJ
white/JJ
(chunk marble/NN mausoleum/NN)
located/VBN
in/IN
(chunk Agra/NNP)
./,
(chunk India/NNP)
./
Commissioned/VBN
by/IN
(chunk Mughal/NNP Emperor/NNP Shah/NNP Jahan/NNP)
in/IN
(chunk memory/NN)
of/IN
his/PRP$
beloved/JJ
(chunk wife/NN)
(chunk Mumtaz/NNP Mahal/NNP)
./,
it/PRP
is/VBZ
an/DT
architectural/JJ
(chunk marvel/NN)
and/CC
a/DT
(chunk UNESCO/NNP World/NNP Heritage/NNP Site/NNP)
./
The/DT
(chunk Taj/NNP Mahal/NNP)
attracts/VBZ
millions/NNS
of/IN
visitors/NNS
each/DT
(chunk year/NN)
who/WP
come/VBP
to/TO
admire/VB
its/PRP$
intricate/JJ
(chunk craftsmanship/NN)
and/CC
timeless/JJ
(chunk beauty/NN)
./.)
```

✓ Practical No 6

Write a program to perform WordNet & also check Word Similarity on English Text.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
[nltk_data] | Downloading package tagsets_json to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package tagsets_json is already up-to-date!
[nltk_data] | Downloading package timit to /root/nltk_data...
[nltk_data] | Package timit is already up-to-date!
[nltk_data] | Downloading package toolbox to /root/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!
[nltk_data] | Downloading package treebank to /root/nltk_data...
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
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[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
True
```

```
from nltk.corpus import wordnet
```

```
syns = wordnet.synsets("program")
print(syns)
print(syns[0])
```

```
[Synset('plan.n.01'), Synset('program.n.02'), Synset('broadcast.n.02'), Synset('platform.n.02'), Synset('program.n.05'), Synset('couple.n.01'), Synset('plan.n.01')]
```

```
print(syns[0].lemmas())
print(syns[0].lemmas()[0].name())
```

```
[Lemma('plan.n.01.plan'), Lemma('plan.n.01.program'), Lemma('plan.n.01.programme')]
plan
```

```
# to find the definition
print(syns[0].definition())
```

```
a series of steps to be carried out or goals to be accomplished
```

```
# to get an example
print(syns[0].examples())
```

```
['they drew up a six-step plan', 'they discussed plans for a new bond issue']
```

```
from nltk.corpus import wordnet
antonyms = []
# defining the function to find Antonyms
def ToFindAntonyms(x):
    for syn in wordnet.synsets(x):
        for lm in syn.lemmas():
            if lm.antonyms():
                antonyms.append(lm.antonyms()[0].name()) #adding into antonyms
    return antonyms
print(set(ToFindAntonyms("bright")))
```

```
{'dull', 'dimmed'}
```

```
print(set(ToFindAntonyms("inactive")))
```

```
{'operational', 'dull', 'dimmed', 'active'}
```

```
print(set(ToFindAntonyms("good")))
```

```
{'operational', 'evilness', 'evil', 'dull', 'active', 'bad', 'badness', 'dimmed', 'ill'}
```

```
synonyms=[]
antonyms=[]
for syn in wordnet.synsets("good"):
    for l in syn.lemmas():
        synonyms.append(l.name())
        if l.antonyms():
            antonyms.append(l.antonyms()[0].name())
print("Synonyms: ",set(synonyms))
print("Antonyms: ",set(antonyms))
```

```
Synonyms: {'dear', 'full', 'goodness', 'right', 'in_effect', 'commodity', 'undecomposed', 'salutary', 'serious', 'good', 'in_force'}
Antonyms: {'evilness', 'evil', 'bad', 'badness', 'ill'}
```

```
# WordSimilarity
car = wordnet.synset('car.n.01')
automobile = wordnet.synset('automobile.n.01')
print("Similarity between car and automobile", car.path_similarity(automobile))
```

```
Similarity between car and automobile 1.0
```

```
# Wu-Palmer Similarity
from nltk.corpus import wordnet as wn
w1 = wordnet.synset('run.v.01') # v here denotes the tag verb
w2 = wordnet.synset('sprint.v.01')
print("Wu-Palmer Similarity between run and sprint", w1.wup_similarity(w2))
```

```
Wu-Palmer Similarity between run and sprint 0.8571428571428571
```

Start coding or [generate](#) with AI.

```
jump = wn.synset('jump.v.01')
leap = wn.synset('leap.v.01')
ship = wn.synset('ship.n.01')
print("Wu-Palmer Similarity between jump and leap", jump.wup_similarity(leap))
```

↗ Wu-Palmer Similarity between jump and leap 1.0

▼ Practical No 7

Write a program to implement word cloud of English Text.

```
from wordcloud import wordcloud
import matplotlib.pyplot as plt

import matplotlib.pyplot as pPlt
from wordcloud import WordCloud, STOPWORDS
import numpy as npy
from PIL import Image

dataset = open("/content/Indian Motorcycle.txt", "r").read()

dataset = dataset.upper()

maskArray = npy.array(Image.open("/content/video-300x300.jpg"))
cloud = WordCloud(background_color = "#000012", min_font_size=5, colormap='Oranges',
                  max_font_size=70, font_path='/content/LEMONMILK-Medium.otf', max_words= 2000, width=1400, height=800,
                  collocations=True, mask = maskArray, stopwords = set(STOPWORDS)).generate(dataset)

# plot the WordCloud image
plt.figure(figsize = (6, 10), facecolor = None)
plt.imshow(cloud)
plt.axis("off")
```

$$\Rightarrow (-0.5, 299.5, 299.5, -0.5)$$


✓ Practical No 8

Write a program to process Text Summarization.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```

[nltk_data] | Package tagsets is already up-to-date!
[nltk_data] | Downloading package tagsets_json to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package tagsets_json is already up-to-date!
[nltk_data] | Downloading package timit to /root/nltk_data...
[nltk_data] | Package timit is already up-to-date!
[nltk_data] | Downloading package toolbox to /root/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!
[nltk_data] | Downloading package treebank to /root/nltk_data...
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_tagset is already up-to-date!
[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True

```

```

import nltk
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
nltk.download('maxent_ne_chunker')
nltk.download('words')

```

```

[ntlk_data] Downloading package punkt to /root/nltk_data...
[ntlk_data] Package punkt is already up-to-date!
[ntlk_data] Downloading package averaged_perceptron_tagger to
[ntlk_data] /root/nltk_data...
[ntlk_data] Package averaged_perceptron_tagger is already up-to-
[ntlk_data] date!
[ntlk_data] Downloading package maxent_ne_chunker to
[ntlk_data] /root/nltk_data...
[ntlk_data] Package maxent_ne_chunker is already up-to-date!
[ntlk_data] Downloading package words to /root/nltk_data...
[ntlk_data] Package words is already up-to-date!
True

```

```
from nltk.tokenize import sent_tokenize, word_tokenize
```

```
dataset = open("/content/Indian Motorcycle.txt", encoding='cp1252').read()
dataset
```

```

'Indian Motorcycle (or Indian) is an American brand of motorcycles owned and produced by American automotive manufacturer Polaris Inc.
Originally produced from 1901 to 1953 in Springfield, Massachusetts, United States, Hendee Manufacturing Company initially produced the motorcycles, but the name was changed to the Indian Motorcycle Company in 1923. In 2011, Polaris Industries purchased the Indian motorcycle marque, moved operations from North Carolina merging them into their existing facilities in Minnesota and Iowa. Since August 2013, Polaris has designed, engineered, and manufactured many lines of motorcycles under the Indian Motorcycle brand reflecting Indian's traditional styling.
The Indian Motorcycle factory team took the first three places in the 1911 Isle of Man TT.

```

```
def summarization(dataset):
```

```

    result = []
    for number, sentence in enumerate(sent_tokenize(dataset)):
        number_tokens = len(word_tokenize(sentence))
        tagged = nltk.pos_tag(word_tokenize(sentence))
        number_nouns = len([word for word, pos in tagged if pos in ['NN', 'NNP']])
        ners = nltk.ne_chunk(nltk.pos_tag(word_tokenize(sentence)), binary=False)
        number_ners = len([chunk for chunk in ners if hasattr(chunk, 'label')])
        score = (number_ners + number_nouns)/float(number_tokens)
        result.append((number, score, sentence))
    return result

```

```
summ = summarization(dataset)
summ
```

```

[(0,
 0.5454545454545454,
 'Indian Motorcycle (or Indian) is an American brand of motorcycles owned and produced by American automotive manufacturer Polaris Inc.')]

```

```
for i in sorted(summ, key=lambda x: x[1], reverse=True):
    print(i[2])
```

```

Indian Motorcycle (or Indian) is an American brand of motorcycles owned and produced by American automotive manufacturer Polaris Inc.

```

✓ Practical No 9

Write a program to implement Word2Vec on Wikipedia Articles and finding the similarity between the words.

```
!pip install wikipedia
```

```
Requirement already satisfied: wikipedia in /usr/local/lib/python3.11/dist-packages (1.4.0)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.11/dist-packages (from wikipedia) (4.13.3)
Requirement already satisfied: requests<3.0.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from wikipedia) (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.0.0->wikipedia) (3.0.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.0.0->wikipedia) (3.0.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.0.0->wikipedia) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests<3.0.0,>=2.0.0->wikipedia) (2024.7.4)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4->wikipedia) (2.6)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4->wikipedia) (4.10.0)
```

```
!pip install gensim
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.11/dist-packages (4.3.3)
Requirement already satisfied: numpy<2.0,>=1.18.5 in /usr/local/lib/python3.11/dist-packages (from gensim) (1.26.4)
Requirement already satisfied: scipy<1.14.0,>=1.7.0 in /usr/local/lib/python3.11/dist-packages (from gensim) (1.13.1)
Requirement already satisfied: smart-open>=1.8.1 in /usr/local/lib/python3.11/dist-packages (from gensim) (7.1.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.11/dist-packages (from smart-open>=1.8.1->gensim) (1.17.2)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] Downloading collection 'all'
[nltk_data] |
[nltk_data] | Downloading package abc to /root/nltk_data...
[nltk_data] | Package abc is already up-to-date!
[nltk_data] | Downloading package alpino to /root/nltk_data...
[nltk_data] | Package alpino is already up-to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger is already up-
[nltk_data] | to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger_eng to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger_eng is already
[nltk_data] | up-to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger_ru to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger_ru is already
[nltk_data] | up-to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger_rus to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger_rus is already
[nltk_data] | up-to-date!
[nltk_data] | Downloading package basque_grammars to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package basque_grammars is already up-to-date!
[nltk_data] | Downloading package bcp47 to /root/nltk_data...
[nltk_data] | Package bcp47 is already up-to-date!
[nltk_data] | Downloading package biocreative_ppi to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package biocreative_ppi is already up-to-date!
[nltk_data] | Downloading package bllip_wsj_no_aux to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package bllip_wsj_no_aux is already up-to-date!
[nltk_data] | Downloading package book_grammars to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package book_grammars is already up-to-date!
[nltk_data] | Downloading package brown to /root/nltk_data...
[nltk_data] | Package brown is already up-to-date!
[nltk_data] | Downloading package brown_te1 to /root/nltk_data...
[nltk_data] | Package brown_te1 is already up-to-date!
[nltk_data] | Downloading package cess_cat to /root/nltk_data...
[nltk_data] | Package cess_cat is already up-to-date!
[nltk_data] | Downloading package cess_esp to /root/nltk_data...
[nltk_data] | Package cess_esp is already up-to-date!
[nltk_data] | Downloading package chat80 to /root/nltk_data...
[nltk_data] | Package chat80 is already up-to-date!
[nltk_data] | Downloading package city_database to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package city_database is already up-to-date!
[nltk_data] | Downloading package cmudict to /root/nltk_data...
```



```
[nltk_data] | Package cmudict is already up-to-date!
[nltk_data] | Downloading package comparative_sentences to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package comparative_sentences is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package comtrans to /root/nltk_data...
[nltk_data] | Package comtrans is already up-to-date!
[nltk_data] | Downloading package conll2000 to /root/nltk_data
```

```
from nltk.tokenize import sent_tokenize
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

import string
import warnings
from gensim.models import Word2Vec
warnings.filterwarnings(action='ignore', category=UserWarning, module='gensim')
```

```
import wikipedia
from wikipedia import search
from wikipedia import page

titles = search("Data Science")
wikipage = page(titles[0])
wikipage.content
wikipage.categories
wikipedia.summary("Data Science", sentences=1)
wikipedia.search("Data Science")
```

```
↗ ['Data science',
  'Data (computer science)',
  'Biomedical data science',
  'Social data science',
  'Data',
  'Data analysis',
  'Master in Data Science',
  'Data type',
  'Data engineering',
  'Branches of science']
```

```
def preprocessing(text):
    result = []
    sent = sent_tokenize(text)
    for sentence in sent:
        words = word_tokenize(sentence)
        tokens = [w for w in words if w.lower() not in string.punctuation]
        stopw = stopwords.words("english")
        tokens = [token for token in tokens if token not in stopw]
        tokens = [word for word in tokens if len(word)>= 3]
        lemma = WordNetLemmatizer()
        tokens = [lemma.lemmatize(word) for word in tokens]
        result += [tokens]

    return result
```

```
text_p = preprocessing(wikipage.content)
text_p[0]
```

```
↗ ['Data',
  'science',
  'interdisciplinary',
  'academic',
  'field',
  'us',
  'statistic',
  'scientific',
  'computing',
  'scientific',
  'method',
  'processing',
  'scientific',
  'visualization',
  'algorithm',
  'system',
  'extract',
  'extrapolate',
  'knowledge',
  'potentially',
  'noisy',
  'structured',
  'unstructured',
  'data']
```

```
# Define your parameters
min_count = 2
vector_size = 50
window = 4

# Create the Word2Vec model
wikimodel = Word2Vec(text_p, min_count=min_count, vector_size=vector_size, window=window)
```

```
# Get the vocabulary
# Error : vocab = list(wikimodel.wv.vocab.keys())

vocab = list(wikimodel.wv.index_to_key)
vocab[:20]
```

```
['data',
 'science',
 'Data',
 'statistic',
 'analysis',
 'information',
 'knowledge',
 'field',
 'computing',
 'The',
 'method',
 'name',
 'domain',
 'computer',
 '==',
 'datasets',
 'scientist',
 'often',
 'learning',
 'statistical']
```

```
wikimodel.wv.most_similar(positive = ["analysis", "field"], topn = 3)
print(wikimodel.wv.similarity("data", "information"))
print(wikimodel.wv.similarity("data", "science"))
print(wikimodel.wv.similarity("data", "computing"))
```

```
0.026082203
0.06795287
-0.054893013
```

```
titles = search("statistic")
wikiage = page(titles[0])
wikiage.content
wikiage.categories
wikipedia.summary("statistic", sentences = 1)
wikipedia.search("statistic")
```

```
['Statistic',
 'T-statistic',
 'Sufficient statistic',
 'Test statistic',
 'Order statistic',
 'Ancillary statistic',
 'U-statistic',
 'Youden's J statistic',
 'Durbin-Watson statistic',
 'K-statistic']
```

```
def preprocessing(text):
    result = []
    sent = sent_tokenize(text)
    for sentence in sent:
        words = word_tokenize(sentence)
        tokens = [w for w in words if w.lower() not in string.punctuation]
        stopw = stopwords.words("english")
        tokens = [token for token in tokens if token not in stopw]
        tokens = [word for word in tokens if len(word)>= 3]
        lemma = WordNetLemmatizer()
        tokens = [lemma.lemmatize(word) for word in tokens]
        result += [tokens]

    return result

text_p = preprocessing(wikiage.content)
text_p[0]
```

```
['Statistics', 'German', 'Statistik', 'orig']
```

```
# Define your parameters
min_count = 2
vector_size = 50
window = 4

# Create the Word2Vec model
wikimodel = Word2Vec(text_p, min_count=min_count, vector_size=vector_size, window=window)
```

```
# Get the vocabulary
vocab = list(wikimodel.wv.index_to_key)
vocab[:20]
```

```
↗ ['statistic',
  'data',
  'statistical',
  'The',
  'hypothesis',
  'probability',
  'study',
  'error',
  'sample',
  'value',
  'method',
  'analysis',
  'variable',
  'population',
  'null',
  'test',
  'Statistics',
  '==',
  'interval',
  'use']
```

```
wikimodel.wv.most_similar(positive = ["statistical", "hypothesis"], topn = 3)
print(wikimodel.wv.similarity("probability", "sample"))
print(wikimodel.wv.similarity("probability", "value"))
print(wikimodel.wv.similarity("data", "error"))
```

```
↗ 0.07949997
0.32219487
0.07644836
```

✓ Practical No 10

Train a model for Movie Review Classification using NLP Techniques.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.4.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
```

```
nltk.download('all')
```

```
[nltk_data] | Package tagsets is already up-to-date!
[nltk_data] | Downloading package tagsets_json to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package tagsets_json is already up-to-date!
[nltk_data] | Downloading package timit to /root/nltk_data...
[nltk_data] | Package timit is already up-to-date!
[nltk_data] | Downloading package toolbox to /root/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!
[nltk_data] | Downloading package treebank to /root/nltk_data...
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_tagset is already up-to-date!
[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] | Done downloading collection all
True
```

```
import random
from nltk.corpus import movie_reviews
```

```
##
```

```
##
```

```
documents = []
for category in movie_reviews.categories():
    for fileid in movie_reviews.fileids(category):
        documents.append((list(movie_reviews.words(fileid)), category))

random.shuffle(documents)
print(documents[1])
```

```
((['scream', '2', 'isn', '', 't', 'quite', 'as', 'clever', 'as', 'its', 'predecessor', ',', 'but', 'it', 'fills', 'the', 'gap', 'wit
```

```
# Normalize the dataset
all_words = []
for w in movie_reviews.words():
    all_words.append(w.lower())

all_words = nltk.FreqDist(all_words)
print(all_words.most_common(15))
print(all_words["love"])
```

```
(((',', 77717), ('the', 76529), (',', 65876), ('a', 38106), ('and', 35576), ('of', 34123), ('to', 31937), ('', 30585), ('is', 25195
```

```
word_features = list(all_words.keys())[:3000]
```

```
def find_features(document):
    words = set(document)
    features = {}
    for w in word_features:
        features[w] = (w in words)
    return features
```

```
print((find_features(movie_reviews.words('neg/cv000_29416.txt'))))
```

```
{'plot': True, ':': True, 'two': True, 'teen': True, 'couples': True, 'go': True, 'to': True, 'a': True, 'church': True, 'party': Tr
```

```
featuresets = [(find_features(rev), category) for (rev, category) in documents]
```

```
training_set = featuresets[:1900]
testing_set = featuresets[1900:]
```

```
classifier = nltk.NaiveBayesClassifier.train(training_set)
```

```
print("Accuracy:",( nltk.classify.accuracy(classifier, testing_set))*100)
```

```
Accuracy: 78.0
```

```
classifier.show_most_informative_features(15)
```

```
Most Informative Features
      sucks = True          neg : pos   =    10.2 : 1.0
      annual = True         pos : neg   =     9.6 : 1.0
      turkey = True         neg : pos   =     7.9 : 1.0
      silverstone = True    neg : pos   =     7.7 : 1.0
      unimaginative = True  neg : pos   =     7.7 : 1.0
      idiotic = True        neg : pos   =     7.5 : 1.0
      schumacher = True     neg : pos   =     7.4 : 1.0
      atrocious = True      neg : pos   =     7.0 : 1.0
      kombat = True         neg : pos   =     7.0 : 1.0
      mena = True           neg : pos   =     7.0 : 1.0
      shoddy = True         neg : pos   =     7.0 : 1.0
      suvari = True         neg : pos   =     7.0 : 1.0
      frances = True        pos : neg   =     7.0 : 1.0
      regard = True         pos : neg   =     7.0 : 1.0
      kidding = True        neg : pos   =     6.4 : 1.0
```